

change in momentum = $p_f - p_i = 2.0 (0.4 - 0.5) = -0.2 \text{ NS}$

- 6 B** By AP, upthrust = weight of fluid displaced. Liquid X has smaller density and hence exerts a smaller upthrust.

Balanced rod implies object P must have a smaller weight and hence smaller mass.

- 7 D** Take moments about pivot